

Square roots and cube roots

A square root is a number that, multiplied by itself once, equals a given number. For example, one square root of 4 is 2, because $2 \times 2 = 4$. Another square root is -2 , as $(-2) \times (-2) = 4$; the square roots of numbers can be either positive or negative. A cube root is a number that, multiplied by itself twice, equals a given number. For example, the cube root of 27 is 3, because $3 \times 3 \times 3 = 27$.

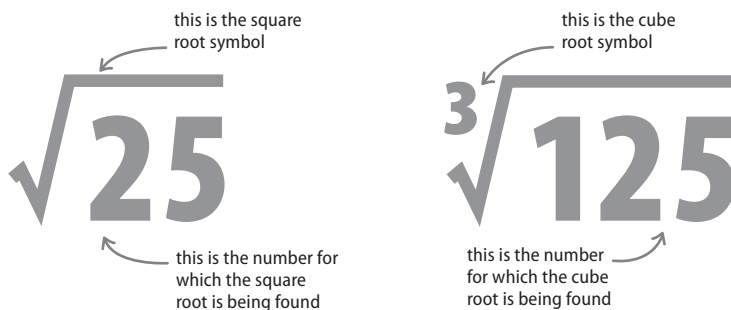


Diagram illustrating the square root of 25: $\sqrt{25} = 5$ because $5 \times 5 = 25$. Labels indicate: 'square root symbol' pointing to the radical sign, 'this is the square root of 25' pointing to the result (5), and '25 is 5^2 ' pointing to the equation $5 \times 5 = 25$.

△ The square root of a number

The square root of a number is the number which, when squared (multiplied by itself), equals the number under the square root sign.

Diagram illustrating the cube root of 125: $\sqrt[3]{125} = 5$ because $5 \times 5 \times 5 = 125$. Labels indicate: 'cube root symbol' pointing to the radical sign, 'this is the cube root of 125' pointing to the result (5), and '125 is 5^3 ' pointing to the equation $5 \times 5 \times 5 = 125$.

△ The cube root of a number

The cube root of a number is the number that, when cubed (multiplied by itself twice), equals the number under the cube root sign.

COMMON SQUARE ROOTS

Square root	Answer	Why?
1	1	Because $1 \times 1 = 1$
4	2	Because $2 \times 2 = 4$
9	3	Because $3 \times 3 = 9$
16	4	Because $4 \times 4 = 16$
25	5	Because $5 \times 5 = 25$
36	6	Because $6 \times 6 = 36$
49	7	Because $7 \times 7 = 49$
64	8	Because $8 \times 8 = 64$
81	9	Because $9 \times 9 = 81$
100	10	Because $10 \times 10 = 100$
121	11	Because $11 \times 11 = 121$
144	12	Because $12 \times 12 = 144$
169	13	Because $13 \times 13 = 169$

LOOKING CLOSER

Using a calculator

Calculators can be used to find powers and square roots. Most calculators have buttons to square and cube numbers, buttons to find square roots and cube roots, and an exponent button, which allows them to raise numbers to any power.



△ Exponent

This button allows any number to be raised to any power.

Diagram illustrating the use of the exponent button: $3^5 = 243$. The calculation is shown as $3 \text{ } X^y \text{ } 5 = 243$.

◁ Using exponents

First enter the number to be raised to a power, then press the exponent button, then enter the power required.



△ Square root

This button allows the square root of any number to be found.

Diagram illustrating the use of the square root button: $25 = 5$. The calculation is shown as $25 \text{ } \sqrt{\quad} = 5$.

◁ Using square roots

On most calculators, find the square root of a number by pressing the square root button first and then entering the number.